

hw 5 solution

T2: (1) 由于 $(x^{-\nu} J_{\nu}(x))' = -x^{-\nu} J_{\nu+1}(x)$ 当 $\nu=0$ 时, 有 $(J_0(x))' = -J_1(x)$

$$\text{令 } ax=t \text{ 则 } \frac{d}{dx} J_0(ax) = \frac{d}{dt} J_0(t) \frac{dt}{dx} = -J_1(t) \cdot a = -a J_1(ax)$$

(2) 由于 $(x^{\nu} J_{\nu}(x))' = x^{\nu} J_{\nu-1}(x)$, 当 $\nu=1$ 时, 有 $(x J_1(x))' = x J_0(x)$

$$\text{令 } ax=t, \text{ 则 } \frac{d}{dx} [x J_1(ax)] = \frac{d}{dt} \left[\frac{t}{a} J_1(t) \right] \frac{dt}{dx} = \frac{1}{a} t J_0(t) \cdot a = ax J_0(ax)$$

$$T3: \text{ 由 } J_{\nu}(x) = \sum_{k=0}^{+\infty} (-1)^k \frac{1}{k! \Gamma(k+\nu+1)} \left(\frac{x}{2}\right)^{2k+\nu} \Rightarrow J_0(x) = \sum_{k=0}^{+\infty} (-1)^k \frac{1}{(k!)^2} \left(\frac{x}{2}\right)^{2k}$$

$$J_0(x \cos \theta) = \sum_{k=0}^{+\infty} (-1)^k \frac{1}{(k!)^2} \left(\frac{x \cos \theta}{2}\right)^{2k}$$

$$\text{故 } \int_0^{\frac{\pi}{2}} J_0(x \cos \theta) \cos \theta d\theta = \int_0^{\frac{\pi}{2}} \sum_{k=0}^{+\infty} (-1)^k \frac{1}{(k!)^2} \left(\frac{x \cos \theta}{2}\right)^{2k} \cos \theta d\theta$$

$$= \sum_{k=0}^{+\infty} (-1)^k \frac{1}{(k!)^2} \left(\frac{x}{2}\right)^{2k} \int_0^{\frac{\pi}{2}} \cos^{2k+1} \theta d\theta = \sum_{k=0}^{+\infty} \frac{(-1)^k}{(k!)^2} \left(\frac{x}{2}\right)^{2k} \frac{(2k)!!}{(2k+1)!!}$$

$$= \sum_{k=0}^{+\infty} \frac{(-1)^k}{(k!)^2} \cdot \left(\frac{x}{2}\right)^{2k} \cdot \frac{2^k \cdot k!}{(2k+1)!!} = \sum_{k=0}^{+\infty} (-1)^k \frac{1}{k!} \frac{x^{2k}}{2^k} \cdot \frac{1}{(2k+1)!!} = \sum_{k=0}^{+\infty} \frac{(-1)^k x^{2k}}{2k!! (2k+1)!!}$$

$$= \sum_{k=0}^{+\infty} \frac{(-1)^k x^{2k}}{(2k+1)!} = \frac{1}{x} \cdot \sin x$$

T5: 例题1的结果为:

$$\cos(x \sin \theta) = J_0(x) + 2[J_2(x) \cos 2\theta + J_4(x) \cos 4\theta + \dots] \quad ①$$

$$\sin(x \sin \theta) = 2[J_1(x) \sin \theta + J_3(x) \sin 3\theta + \dots] \quad ②$$

(1) 令 ① 式中 $\theta=0$

(2) 令 ② 式中 $\theta=\frac{\pi}{2}$

(3) 令 ① 式中 $\theta=\frac{\pi}{2}$

$$T7: (1) \frac{d}{dx} [J_{\nu}^2(x)] = 2 J_{\nu}(x) J'_{\nu}(x) = 2 \frac{x(J_{\nu-1}(x) + J_{\nu+1}(x))}{2\nu} \cdot \frac{J_{\nu-1}(x) - J_{\nu+1}(x)}{2}$$

$$= \frac{x}{2\nu} (J_{\nu-1}^2(x) - J_{\nu+1}^2(x))$$

其中利用公式 $2 J'_{\nu}(x) = J_{\nu-1}(x) - J_{\nu+1}(x)$

$$2\nu x^{-1} J_{\nu}(x) = J_{\nu-1}(x) + J_{\nu+1}(x)$$

$$\begin{aligned}
 (2) \quad \frac{d}{dx} [x J_0(x) J_1(x)] &= J_0(x) J_1(x) + x J_0'(x) J_1(x) + x J_0(x) J_1'(x) \\
 &= J_0(x) J_1(x) - x J_1^2(x) + J_0(x) [x J_0(x) - J_1(x)] \\
 &= J_0(x) J_1(x) - x J_1^2(x) + x J_0^2(x) - J_0(x) J_1(x) \\
 &= x [J_0^2(x) - J_1^2(x)]
 \end{aligned}$$

其中利用公式: $(x J_1(x))' = x J_0(x)$, $J_0'(x) = -J_1(x) = J_{-1}(x)$

T8:

$$\begin{aligned}
 \int_0^x m^n J_0(m) dm &= \int_0^x m^{n-1} (m J_1(m))' dm = \int_0^x m^{n-1} d(m J_1(m)) \\
 &= m^{n-1} \cdot m J_1(m) \Big|_0^x - \int_0^x m J_1(m) (n-1) m^{n-2} dm \\
 &= x^n J_1(x) - (n-1) \int_0^x m^{n-1} J_1(m) dm \\
 &= x^n J_1(x) + (n-1) \int_0^x m^{n-1} J_0'(m) dm \\
 &= x^n J_1(x) + (n-1) \int_0^x m^{n-1} d(J_0(m)) \\
 &= x^n J_1(x) + \left((n-1) m^{n-1} J_0(m) \right) \Big|_0^x - (n-1) \int_0^x J_0(m) (n-1) m^{n-2} dm \\
 &= x^n J_1(x) + (n-1) x^{n-1} J_0(x) - (n-1)^2 \int_0^x m^{n-2} J_0(m) dm
 \end{aligned}$$

$$\begin{aligned}
 (1) \quad \int_0^x x^3 J_0(x) dx &= x^3 J_1(x) + 2x^2 J_0(x) - 4 \int_0^x x J_0(x) dx \\
 &= x^3 J_1(x) + 2x^2 J_0(x) - 4 \int_0^x d(x J_1(x)) \\
 &= x^3 J_1(x) + 2x^2 J_0(x) - 4x J_1(x)
 \end{aligned}$$

$$\begin{aligned}
 (2) \quad \int_0^x x^4 J_1(x) dx &= - \int_0^x x^4 d(J_0(x)) = -x^4 J_0(x) + 4 \int_0^x x^3 J_0(x) dx \\
 &= -x^4 J_0(x) + 4x^3 J_1(x) + 8x^2 J_0(x) - 16x J_1(x) \\
 &= (8x^2 - x^4) J_0(x) + 4(x^3 - 4x) J_1(x)
 \end{aligned}$$

T11:

$$\begin{aligned}
 (1) \quad \int J_0(x) \sin x dx &= x J_0(x) \sin x - \int x d(J_0(x) \sin x) \\
 &= x J_0(x) \sin x - \int x (J_0'(x) \sin x + J_0(x) \cos x) dx
 \end{aligned}$$

$$\int x J_0'(x) \sin x dx + \int x J_0(x) \cos x dx = - \int x J_1(x) \sin x dx$$

$$+ \int \cos x dx J_1(x) = x J_1(x) \cos x + C_1$$

$$\int J_0(x) \sin x dx = x J_0(x) \sin x - x J_1(x) \cos x + C_2$$

$$(2) \int J_0(x) \cos x dx = x J_0(x) \cos x - \int x dJ_0(x) \cos x$$

$$\int x dJ_0(x) \cos x = \int x (J_0'(x) \cos x - J_0(x) \sin x) dx$$

$$= - \int x (\cos x) J_1(x) dx - \int x J_0(x) \sin x dx$$

$$= - \int x J_1(x) \cos x dx - x J_1(x) \sin x + \int x J_1(x) \cos x dx$$

$$= - x J_1(x) \sin x + C_1$$

$$\int J_0(x) \cos x dx = x J_0(x) \cos x + x J_1(x) \sin x + C_2$$

T13:

$$f(x) = \sum_{n=1}^{+\infty} A_n J_1(\omega_n x) \quad \text{因为 } \omega_n \text{ 为 } J_1(x)=0 \text{ 的正实根, 故为第一类边界条件, 且 } \nu=1, a=1$$

$$A_n = \frac{\langle f(x), J_1(\omega_n x) \rangle}{\langle J_1(\omega_n x), J_1(\omega_n x) \rangle} = \frac{\int_0^1 x f(x) J_1(\omega_n x) dx}{\int_0^1 x J_1^2(\omega_n x) dx} = \frac{\int_0^1 x f(x) J_1(\omega_n x) dx}{N_{11}^2}$$

$$= \frac{\int_0^1 x^2 J_1(\omega_n x) dx}{\frac{1}{2} J_2^2(\omega_n)}$$

$$\text{令 } t = \omega_n x$$

$$\int_0^1 x^2 J_1(\omega_n x) dx = \int_0^{\omega_n} \frac{t^2}{\omega_n^3} J_1(t) dt = \frac{1}{\omega_n^3} \omega_n^2 J_2(\omega_n) = \frac{J_2(\omega_n)}{\omega_n}$$

$$\text{故 } A_n = \frac{2}{\omega_n J_2(\omega_n)} \quad \therefore f(x) = \sum_{n=1}^{+\infty} \frac{2}{\omega_n J_2(\omega_n)} J_1(\omega_n x)$$